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Purrfection: How Cats Skillfully Train and Manipulate Humans to Serve  
Their Every Need

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# **Purrfection: How Cats Skillfully Train and Manipulate Humans to Serve Their Every Need**

## **Abstract**

This study explores the intricate ways in which domestic cats employ manipulation tactics to influence human behavior. Utilizing a mixed-methods approach that combines observational data, surveys, and interviews, this research investigates how cats utilize vocalizations, body language, and attention-seeking behaviors to achieve their goals, ranging from soliciting food to initiating play.

The findings reveal that cats predominantly use vocalizations such as meowing, purring, and chirping to manipulate their human companions. Meowing is particularly effective for demanding attention and food, while purring is often used to enhance bonding and comfort. Chirps and trills are employed to encourage play and interaction. Additionally, body language such as kneading, tail positioning, and eye contact play significant roles in communication and manipulation. Attention-seeking behaviors, including climbing, rubbing, and bringing objects, are crucial in eliciting responses from humans.

The research highlights the positive impact of these manipulation tactics on human-cat relationships, although it also acknowledges the potential for frustration and behavioral adjustments by cat owners. The study contributes valuable insights into the complexities of human-animal interactions and suggests pathways for future research, including larger sample sizes, longitudinal studies, and experimental investigations.

This work offers practical implications for enhancing human-cat interactions and improving the understanding of feline behavior, fostering more harmonious relationships between cats and their human companions.

**Keywords:** cat, manipulation, anthrozoology

# Miauestria: Jak koty umiejętnie trenują i manipulują ludźmi w celu zaspokojenia wszystkich swoich potrzeb

## Streszczenie

Niniejsze badanie bada skomplikowane sposoby, w jakie koty domowe stosują taktyki manipulacji, aby wpływać na ludzkie zachowanie. Wykorzystując mieszane podejście metodologiczne, które łączy dane obserwacyjne, ankiety i wywiady, badanie to bada, w jaki sposób koty wykorzystują wokalizacje, język ciała i zachowania poszukujące uwagi, aby osiągnąć swoje cele, od zabiegania o jedzenie po inicjowanie zabawy.

Wyniki badań pokazują, że koty używają głównie wokalizacji, takich jak miauczenie, mruczenie i ćwierkanie, aby manipulować swoimi ludzkimi towarzyszami. Miauczenie jest szczególnie skuteczne w domaganiu się uwagi i pożywienia, podczas gdy mruczenie jest często wykorzystywane w celu zwiększenia więzi i komfortu. Świergoty i tryle są wykorzystywane do zachęcania do zabawy i interakcji. Ponadto język ciała, taki jak ugniatanie, ułożenie ogona i kontakt wzrokowy, odgrywają znaczącą rolę w komunikacji i manipulacji. Zachowania wymagające uwagi, w tym wspinanie się, pocieranie i przynoszenie przedmiotów, mają kluczowe znaczenie dla wywoływania reakcji ze strony ludzi.

Badania podkreślają pozytywny wpływ tych taktyk manipulacji na relacje między człowiekiem a kotem, choć dostrzegają również potencjał frustracji i dostosowania zachowań przez właścicieli kotów. Badanie wnosi cenny wgląd w złożoność interakcji człowiek-zwierzę i sugeruje ścieżki przyszłych badań, w tym większe rozmiary prób, badania podłużne i badania eksperymentalne.

Praca ta oferuje praktyczne implikacje dla poprawy interakcji człowiek-kot i lepszego zrozumienia kocich zachowań, sprzyjając bardziej harmonijnym relacjom między kotami i ich ludzkimi towarzyszami.

**Słowa kluczowe:** algorytm ewolucyjny, optymalizacja, strojenie, hiperparametr, sieć neuronowa

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# 1 Tutorial

**TODO:** Exclude this chapter!

This template uses the New Computer Modern font with size 11pt, a spacing of 1.15 with a side margin of 2.5cm with a .5mm single sided bounding correction (only if you compile it with `in-print` set to `true`), which is *intended* to follow the guidelines of the [Warsaw University of Technology \(WUT\)](#) (this is a named abbreviation, you can find its definition in the `glossary.typ` file). This can be edited in the `thesis.typ` file<sup>1</sup>. If you are looking for how to cite, this sentence is what you are looking for [\[1\]](#) or [J. W. S. Bradshaw \[1\]](#). Note that the citation syntax is the same as the abbreviation syntax and references syntax (more on that later). For more information about citing/bibliography style refer to the [docs](#) and about glossary refer to the [glossary package documentation](#) (note that you might need to update the dependency version in the import statements in `thesis.typ` and `glossary.typ`. This is an abbreviation that is explained in the glossary, which is the *greatest of all time (GOAT)*.

## 1.1 New to Typst?

Here are some resources to help you get on board:

- [Official tutorial](#), highly recommended!
- [Syntax reference](#). In general Typst has good and user friendly reference/documentation
- If you already know a bit about Latex there's a [good migration guide](#)
- [User forum](#) – for when you don't know how to do something and the reference/guides are not enough
- [Repository](#) of user-created packages and templates for typst

## 1.2 Drafting and Printing

Set the boolean variable `draft` or `in-print` in the `thesis.typ`.

The `draft` variable is used to show TODOs, change the coloring of some elements, and DRAFT in the header and the title. It should be `true` until the final version is handed-in.

The `in-print` variable is a boolean:

- set to `true` to generate a PDF file for **physical printing** (it adds bigger margins on the binding part, changes the numbering of the page and chapter headers from centered to print-sided ones and turns off link coloring);
- set to `false` to create a PDF file for **reading on screen** (e.g. to upload to onedrive for the **final thesis hand-in**).

---

<sup>1</sup>This is a footnote.

### 1.3 Choosing a correct faculty

To set a correct abbreviation in the `faculty` argument in `thesis.typ` consult the [Table 1](#). Note that this table is generated programmatically (using the `faculty` dictionary, but you can achieve similar things by loading data files).

**Table 1.** Faculties and their abbreviations to use in the settings of the thesis

| Abbr.  | Polish name                                           | English name                                                       |
|--------|-------------------------------------------------------|--------------------------------------------------------------------|
| kneis  | Kolegium Nauk Ekonomicznych i Społecznych w Płocku    | College of Economics and Social Sciences (Plock)                   |
| wains  | Wydział Administracji i Nauk Społecznych              | Faculty of Administration and Social Sciences                      |
| warch  | Wydział Architektury                                  | Faculty of Architecture                                            |
| wbmip  | Wydział Budownictwa, Mechaniki i Petrochemii w Płocku | Faculty of Civil Engineering, Mechanics and Petrochemistry (Plock) |
| wch    | Wydział Chemiczny                                     | Faculty of Chemistry                                               |
| we     | Wydział Elektryczny                                   | Faculty of Electrical Engineering                                  |
| weiti  | Wydział Elektroniki i Technik Informacyjnych          | Faculty of Electronics and Information Technology                  |
| wf     | Wydział Fizyki                                        | Faculty of Physics                                                 |
| wgik   | Wydział Geodezji i Kartografii                        | Faculty of Geodesy and Cartography                                 |
| wichip | Wydział Inżynierii Chemicznej i Procesowej            | Faculty of Chemical and Process Engineering                        |
| wil    | Wydział Inżynierii Lądowej                            | Faculty of Civil Engineering                                       |
| wim    | Wydział Inżynierii Materiałowej                       | Faculty of Materials Science and Engineering                       |
| wis    | Wydział Inżynierii Środowiska                         | Faculty of Environmental Engineering                               |
| wmeil  | Wydział Mechaniczny Energetyki i Lotnictwa            | Faculty of Power and Aeronautical Engineering                      |
| wmini  | Wydział Matematyki i Nauk Informacyjnych              | Faculty of Mathematics and Information Science                     |
| wmchtr | Wydział Mechatroniki                                  | Faculty of Mechatronics                                            |
| wmt    | Wydział Mechaniczny Technologiczny                    | Faculty of Mechanical and Industrial Engineering                   |
| wsimr  | Wydział Samochodów i Maszyn Roboczych                 | Faculty of Automotive and Construction Machinery Engineering       |
| wt     | Wydział Transportu                                    | Faculty of Transport                                               |
| wz     | Wydział Zarządzania                                   | Faculty of Management                                              |

### 1.4 Example figures

#### 1.4.1 Normal Figure

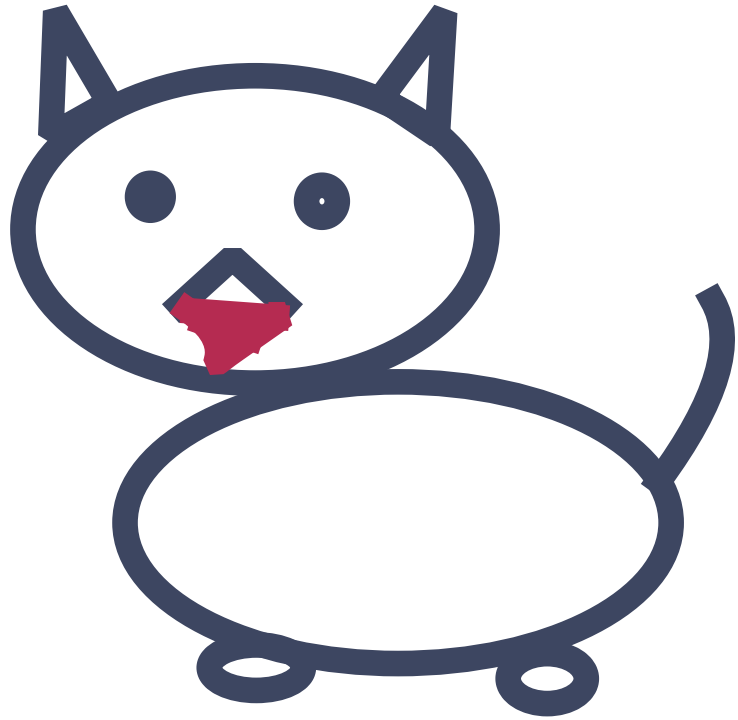
The reference system works using the same syntax for both citations and references to document's elements. For example this is a reference to [Figure 1](#). And this is a reference to this chapter: [Chapter 1](#). And this is a reference to a label from a different chapter: [Appendix A](#) (it just works). If you want to customize the supplement text you can do so in such a way: “W [Rozdziale 1](#) znajduje się poradnik”.



**Figure 1.** This is a caption beautiful cat named Miss Moneyppenny

### 1.4.2 Wrap text around image

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aequaleam animo, cum corpore doleamus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre in voluptatem, ut postea variari voluptas distinguique possit, augeri amplificarique non possit. At etiam Athenis, ut e patre audiebam facete et urbane Stoicos irridente, statua est in quo a nobis philosophia defensus et collaudata est, cum id, quod maxime placeat, facere pos-



**Figure 2.** another caption

simus, omnis voluptas assumenda est, omnis dolor repellendus. Temporibus autem quibusdam et.

### 1.4.3 Sub-Figures

To achieve referencable subfigures it might be easier to use the [subpar package](#).



a) test 1



b) test2

**Figure 3.** Caption

## 1.5 Math

This is how you define inline math:  $a^2 + b^2 = c^2$ . For bigger equations you can use a math block, as you can see in [Equation 1](#).

$$\int_{-\infty}^{\infty} f(x) dx = 1 \quad (1)$$

$$1 = 1$$

For more math info see [docs](#).

## 1.6 Tables

If you want to style tables, I recommend this [in-depth table guide](#).

### 1.6.1 Normal Table

| Substance             | Subcritical °C | Supercritical °C |
|-----------------------|----------------|------------------|
| Hydrochloric Acid     | 12.0           | 92.1             |
| Sodium Myreth Sulfate | 16.6           | 104              |
| Potassium Hydroxide   | 24.7           |                  |

### 1.6.2 Referenceable Table with a caption

**Table 2.** This is a caption

| Substance             | Subcritical °C | Supercritical °C |
|-----------------------|----------------|------------------|
| Hydrochloric Acid     | 12.0           | 92.1             |
| Sodium Myreth Sulfate | 16.6           | 104              |
| Potassium Hydroxide   | 24.7           |                  |

And this is a reference to [Table 2](#).

**This is a silent level 4 heading**

This won't show up in the overview.

## 1.7 Code

You can create listings from the embedded code like so:

**Listing 1.** Hello world in the Python language

```
print("Hello World")
```

You can also load listings directly from source files, as in the [Listing 2](#).

**Listing 2.** Collatz sequence

---

```
struct Collatz {
    current_value: u32,
}

impl Collatz {
    fn new(initial_value: u32) -> Self {
        Collatz {
            current_value: initial_value,
        }
    }

    fn update_val(&mut self) {
        if self.current_value % 2 == 0 {
            self.current_value /= 2;
        } else {
            self.current_value = self.current_value * 3 + 1;
        }
    }

    fn print_sequence(&mut self) {
        let mut i = 1;
        while self.current_value > 1 {
            println!("value {} = {}", i, self.current_value);
            self.update_val();
            i += 1;
        }
    }
}

fn main() {
    // prints Collatz sequence, starting from 194375
    let mut seq = Collatz::new(194_375);
    seq.print_sequence();
}
```

---

If you want to change the theme of the syntax highlighting you can set the `raw.theme` parameter. If you want to further customize the look of the code blocks you might want to use a package like `codly` or similar.

## 1.8 Algorithms

In this example I use `lovelace` package to typeset the algorithm but there are several other packages which provide similar functionality. Algorithms are fully referenceable: [Algorithm 1](#)

**Algorithm 1.** The infamous Bogosort algorithm

---

**Require:** *deck* is an unsorted array of integers

```
1 function BOGOSORT(deck) ▷ Optimized for speed
2   while deck is not sorted do
3     ⊣ SHUFFLE(deck)
4   ⊣ return deck
```

---

## 2 Introduction

**TODO:** replace this chapter with your introduction

### 2.1 Background and Motivation

Cats have been companions to humans for thousands of years, yet their unique behavior and the subtle art of their interaction with humans remain only partially understood. Unlike dogs, whose domestication has long been associated with clear roles in human societies, cats have carved out a distinct niche within domestic settings. Their autonomy, coupled with an apparent mastery of human psychology, positions them as enigmatic figures in the world of pets [1, p.20]. This peculiar relationship motivates the current study, which seeks to explore how cats manage to manipulate and train their human companions to cater to their needs and desires.

Feline behavior is often characterized by an apparent nonchalance and independence, yet this exterior masks a sophisticated interplay of manipulation and affection [2, pp. 90-110]. Observations indicate that cats use a variety of subtle and overt strategies to influence human behavior, from purring and meowing to more nuanced actions like “kneading” and selective attention-seeking [3]. This complex interaction raises questions about the cognitive and emotional dynamics at play and how these influence both the cats’ and their human caregivers’ daily lives.

### 2.2 Research Gap

While substantial research exists on domestic animals and their roles in human lives, the focus has predominantly been on dogs, often highlighting their roles as companions, workers, or service animals. In contrast, the behavioral intricacies of cats and their impact on human behavior are less thoroughly explored [4]. Most studies in feline behavior have concentrated on health-related issues or general behavioral patterns, leaving a significant gap in understanding the specific ways in which cats manipulate their human counterparts.

Current literature does not fully address how cats employ particular behaviors to influence human actions or how these interactions affect human psychology and household dynamics [5]. The existing research often lacks a detailed analysis of the behavioral strategies employed by cats to achieve their goals, such as gaining attention, securing food, or ensuring comfort [6]. This thesis aims to address these gaps by providing a comprehensive examination of feline manipulation tactics and their consequences for human behavior.

### 2.3 Research Objectives and Questions

The primary aim of this thesis is to delve into the ways cats influence and manipulate their human companions. This exploration involves identifying and categorizing the specific behaviors that cats use to affect human actions, understanding the psychological mechanisms that underlie these behaviors, and assessing the broader impact on human routines and emotional well-being.

To achieve these objectives, the research will address several key questions. First, it will investigate the precise behaviors employed by cats to manipulate their human caregivers.

This includes examining how various tactics, such as vocalizations, body language, and other interactions, are strategically used to achieve specific goals. Secondly, the study will explore the cognitive and emotional processes involved in these manipulation tactics, seeking to understand the underlying psychological mechanisms that enable cats to exert influence over humans. Finally, the research will evaluate the effects of feline behaviors on human behavior, including changes in daily routines, emotional responses, and overall household dynamics.

By addressing these questions, the thesis aims to provide a comprehensive understanding of feline manipulation and its implications for both cats and their human companions.

## 3 Literature Review

**TODO:** Replace this chapter!

### 3.1 Overview of Feline Behavior

Feline behavior has been a subject of interest in animal behavior research, though often overshadowed by studies focusing on dogs. Research has highlighted the domestic cat's unique blend of independence and social interaction. Key studies have documented general behaviors such as purring, kneading, and scratching, as well as their implications for cat health and well-being [1]. However, these studies often overlook the more nuanced aspects of how these behaviors impact human interactions.

### 3.2 Human-Animal Interactions

The field of human-animal interactions has provided valuable insights into the bond between pets and their owners. Research shows that animals, including cats, can influence human emotions and behaviors through various mechanisms [6]. Studies on dogs have revealed that they can alter human social interactions and emotional states, suggesting similar potential for cats. However, the specific ways in which cats employ behavioral strategies to manipulate human behavior have not been thoroughly examined [5].

### 3.3 Comparative Studies: Cats vs. Dogs

Comparative studies between cats and dogs reveal differences in their interactions with humans. Dogs, as pack animals, tend to engage in more overtly cooperative and submissive behaviors, which are well-documented and studied. In contrast, cats, as solitary hunters, exhibit a more subtle and independent approach to their interactions [4]. While dogs have been shown to use behaviors such as eye contact and body language to solicit attention and cooperation, cats often rely on more covert tactics, such as strategic vocalizations and selective affection [3].

### 3.4 Theories of Animal Manipulation

Theories of animal manipulation suggest that animals use various strategies to influence human behavior. For example, the “manipulative attachment theory” posits that pets use attachment behaviors to elicit caregiving responses from their owners. In cats, these behaviors might include specific vocalizations, body movements, and patterns of attention-seeking [2]. However, existing research primarily focuses on the general impact of these behaviors rather than the specific strategies cats use to achieve their goals.

### 3.5 Gaps in the Literature

Despite the considerable research on domestic animals, significant gaps remain in understanding the precise ways in which cats manipulate their human companions. While there is some research on general feline behavior, there is limited investigation into how these behaviors are used

strategically to influence human actions [5]. The subtle nature of feline manipulation, combined with the focus on more overt animal behaviors, has resulted in a lack of comprehensive studies examining the detailed tactics cats use and their effects on human behavior and household dynamics.

## 4 Methodology

**TODO:** Replace this chapter!

### 4.1 Research Design

This study employs a mixed-methods approach to comprehensively investigate feline manipulation tactics and their effects on human behavior. By combining qualitative and quantitative methods, the research aims to provide a detailed understanding of the specific behaviors cats use to influence their human companions, as well as the psychological and emotional impacts of these behaviors.

### 4.2 Participants

#### 4.2.1 Feline Subjects

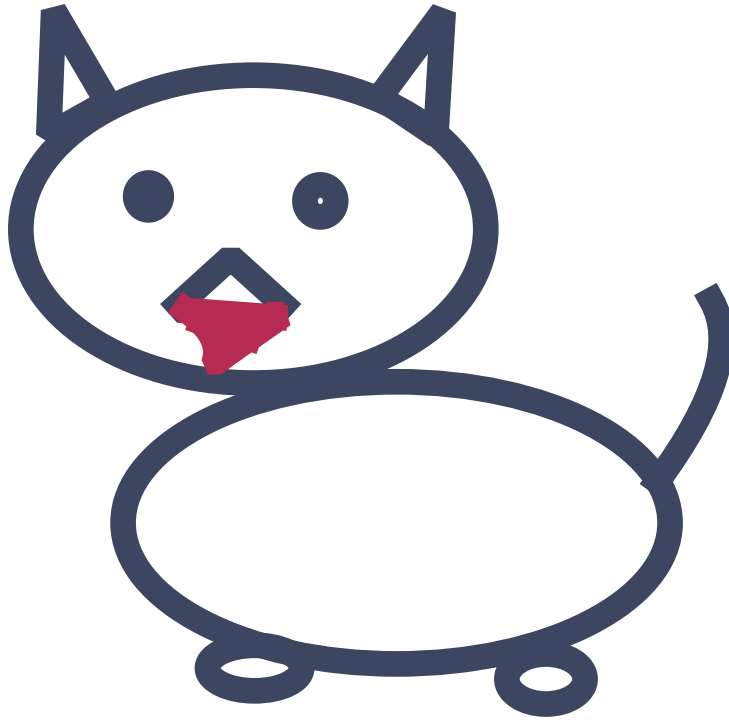
The study includes a sample of domestic cats from diverse backgrounds to ensure a broad understanding of feline manipulation tactics. Cats are selected based on criteria such as age, breed, and living conditions, with the goal of representing a variety of typical domestic scenarios. The following images are examples of the participating cats:

- Miss Moneypenny: female, ragdoll, 11 years



**Figure 4.** Beautiful cat named Miss Moneypenny

- James Bond: male, ragdoll, 11 years



**Figure 5.** Charming cat named James Bond

- Calcipher: male, oriental short hair, 4 months



**Figure 6.** Adorable kitten named Calcipher

- Mei: female, oriental short hair, 4 months



**Figure 7.** Cute kitten named Mei

#### 4.2.2 Human Subjects

Human participants are recruited from households with cats to observe interactions and gather insights into the impact of feline behavior. Inclusion criteria include having lived with the cat for at least six months to ensure familiarity with the cat's behavioral patterns. Participants are selected to represent a range of demographics to capture diverse experiences.

### 4.3 Data Collection Methods

#### Observational Study

Observations are conducted in naturalistic settings to document feline behaviors and their impact on human responses. Cats and their human companions are observed over a period of six weeks, focusing on interactions related to manipulation tactics such as vocalizations, body language, and attention-seeking behaviors. Observations are recorded using video and field notes to capture detailed instances of manipulation. Surveys and Interviews

Surveys are distributed to human participants to gather quantitative data on their perceptions of feline behavior and its effects on their daily routines and emotional well-being. The survey includes questions about the frequency of specific behaviors, perceived manipulative tactics, and changes in human behavior as a result.

In-depth interviews are conducted with a subset of participants to explore their experiences and insights in greater detail. Interviews focus on specific instances of feline manipulation and its impact on household dynamics and personal feelings. The interviews are semi-structured, allowing flexibility to explore relevant topics that arise during the conversation.

## **4.4 Data Analysis**

### **4.4.1 Qualitative Analysis**

Data from observations and interviews are analyzed using thematic analysis to identify common patterns and themes in feline manipulation tactics and human responses. Key behaviors and strategies are categorized, and the relationships between these behaviors and human reactions are explored.

#### **4.4.1.1 Quantitative Analysis**

Survey data are analyzed using statistical techniques to quantify the prevalence of specific behaviors and their perceived impact. Descriptive statistics, such as frequency distributions and measures of central tendency, are used to summarize the data. Inferential statistics, such as correlation and regression analyses, are employed to examine relationships between feline behaviors and changes in human routines and emotions.

## **4.5 Ethical Considerations**

Ethical approval is obtained from the relevant institutional review board (IRB) to ensure that all research activities adhere to ethical standards. Informed consent is obtained from all human participants, and confidentiality is maintained by anonymizing personal data. Feline subjects are observed in their natural environment without any intervention that could affect their well-being.

## **4.6 Limitations**

The study acknowledges potential limitations, including the observational nature of the research, which may introduce observer bias. Additionally, the sample size for both feline and human participants may limit the generalizability of the findings. Efforts are made to minimize these limitations by employing rigorous data collection and analysis methods.

## 5 Data Collection

**TODO:** Replace this chapter!

This chapter details the data collection process used to investigate feline manipulation tactics and their impact on human behavior. The study utilized a combination of observational methods, surveys, and interviews to gather comprehensive data on feline behaviors and human responses. This multi-method approach was designed to provide a well-rounded understanding of how cats influence their human companions.

### 5.1 Observational Study

#### 5.1.1 Procedure

Observations were conducted in participants' homes to capture naturalistic interactions between cats and their human caregivers. The study took place over a six-week period, during which various behaviors were recorded to understand manipulation tactics [5].

#### 5.1.2 Data Collection Tools

- **Video Recording:** Cameras were placed in key areas of the home to capture feline behaviors unobtrusively. Video recordings allowed for detailed analysis of vocalizations, body language, and attention-seeking actions [4].
- **Field Notes:** Observers took detailed notes to complement video recordings. These notes provided context and insights into behaviors that were not always captured on camera [2].

#### 5.1.3 Focus Areas

- **Vocalizations:** Different types of vocalizations, such as meows, purrs, and chirps, were documented. Observers noted the frequency, pitch, and context of these vocalizations [3].
- **Body Language:** Behaviors such as kneading, tail positioning, and eye contact were observed and categorized. Attention was given to how these behaviors were used to elicit specific responses from humans [1].
- **Attention-Seeking Behaviors:** Actions like climbing on furniture, rubbing against humans, and bringing objects were recorded to assess their effectiveness in gaining attention or achieving desired outcomes [6].

### 5.2 Surveys

#### 5.2.1 Design

Surveys were developed to collect quantitative data on human perceptions of feline behavior and its effects. The survey included questions on:

- **Frequency of Behaviors:** How often participants observed specific manipulation tactics.
- **Effectiveness:** How effective participants believed these behaviors were in achieving certain outcomes, such as getting fed or receiving attention.

- **Impact on Routines and Emotions:** How these behaviors affected participants' daily routines and emotional states.

### 5.2.2 Distribution

Surveys were distributed to human participants through both electronic and paper formats. The survey was designed to be accessible and easy to complete. Follow-up reminders were sent to ensure a high response rate and comprehensive data collection [4].

### 5.2.3 Analysis

Survey responses were analyzed to identify trends and patterns in human perceptions of feline manipulation. The data provided insights into the frequency of specific behaviors and their perceived effectiveness in influencing human behavior [2].

## 5.3 Interviews

### 5.3.1 Procedure

In-depth interviews were conducted with a subset of survey participants to gather qualitative insights into their experiences. Interviews were semi-structured, allowing for exploration of specific instances and personal reflections [1].

### 5.3.2 Topics Covered

- **Instances of Manipulation:** Participants described notable examples where their cats' behaviors led to specific outcomes.
- **Emotional Responses:** Participants discussed their emotional reactions to their cats' manipulative tactics.
- **Behavioral Changes:** Insights were gathered on how participants' routines and attitudes towards their cats were affected [5].

### 5.3.3 Data Collection

Interviews were recorded and transcribed for analysis. The transcriptions were coded to identify common themes and patterns related to feline manipulation tactics and their impact on human behavior [6].

## 6 Analysis

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### 6.1 Introduction

The analysis aims to identify patterns and relationships within the data, offering insights into how cats use specific behaviors to influence their human companions and the subsequent effects on human routines and emotions.

### 6.2 Analysis of Observational Data

#### 6.2.1 Behavioral Patterns

##### 6.2.1.1 Vocalizations

The observational data revealed distinct patterns in feline vocalizations used for manipulation:

- **Meowing:** Cats primarily used meowing to solicit food or attention. High-pitched and frequent meows were associated with feeding times, and variations in pitch were observed to correspond with different types of requests.
- **Purring:** Cats employed purring as a multifaceted strategy to seek comfort or attention. Instances of purring were often accompanied by other behaviors, such as rubbing or kneading, reinforcing its role in manipulation.
- **Chirps and Trills:** These vocalizations were less frequent but used effectively to prompt interaction or play, especially when the cat was engaged with a toy or seeking active play.

##### 6.2.1.2 Body Language

Key body language strategies observed included:

- **Kneading:** Cats used kneading predominantly to solicit attention and comfort. This behavior was frequently observed on human laps or soft surfaces, and was often followed by increased physical interaction from humans.
- **Tail Positioning:** The tail's position served as an indicator of the cat's intent and emotional state. Raised tails were associated with positive interactions, while low or flicking tails often preceded withdrawal or irritation.
- **Eye Contact:** Slow blinking and direct eye contact were employed to establish trust and encourage affection. Cats that used these behaviors more frequently were often rewarded with petting or close contact from their humans.

##### 6.2.1.3 Attention-Seeking Behaviors

Common attention-seeking behaviors included:

- **Climbing and Jumping:** Cats used climbing and jumping to place themselves physically in the human's space, often leading to increased interaction or feeding.

- **Rubbing and Head-Butting:** These behaviors were consistently used to solicit affection or attention. Cats that engaged in these actions more frequently were noted to receive more positive responses from their owners.
- **Bringing Objects:** Cats that brought toys to their humans effectively initiated play sessions, demonstrating the use of object-oriented behaviors to engage their caregivers.

## 6.3 Analysis of Survey Data

### 6.3.1 Frequency and Effectiveness

#### 6.3.1.1 Frequency of Behaviors

Survey responses indicated that vocalizations, particularly meowing, were the most commonly observed manipulation tactic. Purring and attention-seeking behaviors such as climbing and rubbing were also reported frequently, with variations based on the cat's individual personality and the household environment.

#### 6.3.1.2 Perceived Effectiveness

Participants rated meowing and purring as the most effective behaviors for soliciting attention and food. Attention-seeking actions like climbing and object bringing were perceived as effective in prompting immediate interaction or play. The effectiveness of these behaviors was found to correlate with the frequency of their occurrence and the specific context in which they were used.

## 6.4 Analysis of Interview Data

### 6.4.1 Thematic Insights

#### 6.4.1.1 Instances of Manipulation

Interviews revealed detailed accounts of how cats used specific behaviors to achieve desired outcomes. Common examples included cats meowing persistently before feeding time and using purring to calm their humans during stressful moments.

#### 6.4.1.2 Emotional Reactions

Participants described a range of emotional responses to their cats' manipulation tactics, from amusement and affection to occasional frustration. The manipulation tactics often led to increased bonding and interaction, though some participants reported feeling manipulated or pressured by their cats' behaviors.

#### 6.4.1.3 Behavioral Changes

The interviews highlighted changes in human behavior in response to feline manipulation. Many participants adjusted their routines to accommodate their cats' needs, such as altering feeding schedules or engaging in more playtime. These changes were generally perceived as positive, contributing to a stronger bond between the cat and its owner.

The analysis of observational, survey, and interview data collectively demonstrates that cats employ a variety of manipulation tactics to influence their human companions. Vocalizations,

body language, and attention-seeking behaviors are used strategically to achieve specific outcomes. The effectiveness of these tactics varies based on the context and the individual cat's behavior. Human responses to these manipulative behaviors are generally positive, though they can occasionally lead to feelings of frustration or manipulation.

## 7 Conclusion

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### 7.1 Summary of Findings

This thesis investigated feline manipulation tactics and their impact on human behavior through a multi-method approach involving observational studies, surveys, and interviews. The key findings of this research can be summarized as follows:

- **Vocalizations:** Cats use a variety of vocalizations, including meows, purrs, and chirps, to manipulate human behavior. Meowing, particularly when high-pitched and frequent, was primarily used to solicit food or attention. Purring was employed both to seek comfort and to reinforce affectionate interactions. Chirps and trills were less common but effectively used to prompt play and interaction.
- **Body Language:** Body language plays a crucial role in feline manipulation. Kneading, tail positioning, and eye contact were observed to be significant in influencing human responses. Kneading was used to solicit attention and comfort, while tail positioning indicated the cat's emotional state and intent. Slow blinking and direct eye contact helped build trust and encouraged affection.
- **Attention-Seeking Behaviors:** Cats engage in various attention-seeking behaviors such as climbing on furniture, rubbing against humans, and bringing objects to prompt interaction. These behaviors were effective in gaining attention and initiating play, often leading to increased human engagement and positive interactions.
- **Human Responses:** Survey and interview data revealed that humans generally respond positively to feline manipulation tactics. Many participants reported adjusting their routines and behaviors to accommodate their cats' needs. While these manipulative behaviors often led to enhanced bonding and affection, some participants also experienced frustration or a sense of being manipulated.

### 7.2 Implications of Findings

The findings of this study have several implications for understanding the dynamics of human-feline relationships:

- **Understanding Feline Behavior:** By identifying and categorizing feline manipulation tactics, this research provides valuable insights into how cats influence human behavior. This understanding can help cat owners recognize and interpret their pets' needs and motivations more effectively.
- **Human-Cat Interaction:** The study highlights the complex nature of human-cat interactions and the role of manipulation in strengthening the bond between cats and their owners. Recognizing and responding to feline manipulation can lead to more harmonious and satisfying relationships.
- **Behavioral Adjustments:** The data suggests that while cats' manipulation tactics can lead to positive outcomes, they may also necessitate changes in human routines and behaviors.

Being aware of these dynamics can help owners manage their interactions with their cats more consciously.

### 7.3 Limitations of the Study

Several limitations were noted during the research:

- **Sample Size:** The study's sample size was limited, which may affect the generalizability of the findings. A larger and more diverse sample could provide a broader perspective on feline manipulation tactics.
- **Observational Bias:** Observations were conducted in naturalistic settings, which may have introduced observational bias. While efforts were made to minimize this, the presence of observers could have influenced the cats' behaviors.
- **Self-Reporting Bias:** Survey and interview data were subject to self-reporting bias, as participants' perceptions and recollections may not always accurately reflect their experiences.

### 7.4 Recommendations for Future Research

To build on the findings of this study, future research could consider the following recommendations:

- **Larger Sample Sizes:** Including a larger and more varied sample of cats and human participants can enhance the generalizability of the results and provide a more comprehensive understanding of feline manipulation tactics.
- **Longitudinal Studies:** Conducting longitudinal studies could offer insights into how feline manipulation tactics and human responses evolve over time and in different life stages of both cats and their owners.
- **Experimental Research:** Experimental studies could explore the causal relationships between specific feline behaviors and human responses, providing a more detailed understanding of how manipulation tactics affect human behavior.

### 7.5 Final Thoughts

This research contributes to a deeper understanding of the complex interactions between cats and their human companions. By examining the manipulation tactics employed by felines and their effects on human behavior, the study sheds light on the nuanced ways in which cats influence their environment. The insights gained from this research not only enhance our knowledge of feline behavior but also offer practical guidance for improving human-cat relationships.

## Acknowledgements

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## List of Symbols and Abbreviations

*GOAT* – greatest of all time

*KDE Community*: An international team developing and distributing Open Source software.

*OIDC* – **OpenID Connect**: OpenID is an open standard and decentralized authentication protocol promoted by the non-profit [OpenID Foundation](#).

*WUT* – **Warsaw University of Technology**

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## Appendix A Supplementary Material

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### Introduction

Thank you for participating in this survey. Your responses will help us understand how feline manipulation tactics affect human behavior. Please answer the following questions based on your experiences with your cat(s).

#### 1. Frequency of Observed Behaviors

- How often does your cat meow to get your attention?
  - ☐ Rarely
  - ☐ Occasionally
  - ☐ Frequently
  - ☐ Very frequently
- How often does your cat purr to get your attention?
  - ☐ Rarely
  - ☐ Occasionally
  - ☐ Frequently
  - ☐ Very frequently
- How often does your cat bring objects (e.g., toys) to you?
  - ☐ Rarely
  - ☐ Occasionally
  - ☐ Frequently
  - ☐ Very frequently

#### 2. Effectiveness of Behaviors

- How effective do you find meowing in getting your cat's needs met?
  - ☐ Not effective
  - ☐ Slightly effective
  - ☐ Moderately effective
  - ☐ Very effective
- How effective do you find purring in calming or eliciting a positive response from you?
  - ☐ Not effective
  - ☐ Slightly effective
  - ☐ Moderately effective
  - ☐ Very effective
- How effective is bringing objects (e.g., toys) in initiating play sessions or interaction?

- ☐ Not effective
- ☐ Slightly effective
- ☐ Moderately effective
- ☐ Very effective

3. Impact on Routines and Emotions

- How has your cat's behavior affected your daily routine?
  - ☐ No impact
  - ☐ Minor impact
  - ☐ Moderate impact
  - ☐ Significant impact
- How do you generally feel when your cat uses manipulation tactics?
  - ☐ Frustrated
  - ☐ Amused
  - ☐ Indifferent
  - ☐ Affectionate

4. Additional Comments

- Please provide any additional comments or experiences related to your cat's manipulation tactics.

## Appendix B Interview Guide

### Introduction

Thank you for agreeing to participate in this interview. The following questions are designed to gather detailed insights into your experiences with your cat's behavior. Feel free to elaborate on your answers as much as you like.

1. Instances of Manipulation

- Can you describe a specific instance where your cat used manipulation tactics to achieve something?
- How did you respond to this behavior?

2. Emotional Responses

- How did you feel during and after the manipulation?
- Did the behavior change your emotional state or relationship with your cat?

3. Behavioral Changes

- Have you made any changes to your routine or behavior in response to your cat's manipulation?
- How have these changes affected your interactions with your cat?

4. General Impressions

- What do you think about the manipulation tactics used by your cat?
- Are there any particular behaviors you find especially effective or challenging?

5. This is a cat:



**Figure A.1.** Example of a cat

6. This is a table:

Table A.1. Example of a table

| Team member     | Monday   | Tuesday | Wednesday | Thursday | Friday |
|-----------------|----------|---------|-----------|----------|--------|
| Evelyn Archer   | Office   |         | Remote    |          | Office |
| Lila Montgomery | On leave |         |           |          |        |
| Nolan Pearce    | Remote   | Office  |           | Remote   | Office |

## TODOs

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